THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 10: **The Other Side of Zero** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** -13	Add step by step.
2	**(d)** 8	Add the opposites.
3	**(a)** 0	Opposites cancel.
4	**(a)** 17	Highest minus lowest.
5	**(b)** -30	Subtract the descent.
6	**(a)** -7 and 12	Check each pair.
7	**(a)** 4	Count the positive integers up to b-1.
8	**(a)** -3	Add step by step.
9	**(d)** -10	Add the opposites.
10	**(a)** 0	Opposites cancel.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Add the first two: 7 (1); Then add (-9): -2 (1).	1+1
12	$-6-9=-15$ (1); $9-(-6)=15$ (1).	1+1
13	Integers: -4, -3, -2, -1, 0, 1 (1); Number $=6$ (1).	1+1
14	Add the first two: -15 (1); Then add 12: -3 (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Start at 0; after 3 down: -3; After 5 up: 2 (1); After 4 down: -2 (1); Total floors moved $=12$ (1).	1+1+1
16	After first: 100; After second: -150 (1); After deposit: -30 (1); Negative means overdrawn (1).	1+1+1
17	$6-(-4)=10$ (1); $10+(-9)=1$ (1); $1-5=-4$ (1).	1+1+1
	OR: Start at 0; after 6 down: -6; After 2 up: -4 (1); After 7 down: -11 (1); Total floors moved $=15$ (1).	1+1+1
18	After first: 50; After second: -30 (1); After deposit: 30 (1); Positive balance (1).	1+1+1
	OR: $-8-3=-11$ (1); $-11+(-7)=-18$ (1); $-18-(-2)=-16$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	After first jump: 5 (1); After second: 0 (1); After third: -9 (1); Distance from 0 $=9$ (2).	1+1+1+2
	OR: Coldest to hottest: -8, -5, -2, 3, 11 (1); Hottest 11, coldest -8 (1); Difference $=19$ (1); Sum $=-1$ (2).	1+1+1+2
20	After first jump: 8 (1); After second: -2 (1); After third: -6 (1); Distance from 0 $=6$ (2).	1+1+1+2
	OR: Coldest to hottest: -12, -9, 0, 6, 14 (1); Hottest 14, coldest -12 (1); Difference $=26$ (1); Sum $=-1$ (2).	1+1+1+2

Mistakes to avoid:

- Saying $-7>-3$ because 7 is greater than 3; on the number line -7 lies to the left of -3.
- Writing $-4-(-6)=-10$; the correct value is $-4+6=2$.
- Dropping the sign of the bigger number when adding integers with different signs, e.g. $(-9)+5=4$ instead of -4.
- Taking the successor of a negative integer as one smaller (successor of -1 is 0, not -2).
- Finding a fall or rise as initial − final in the wrong order; use final − initial and read the sign.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 10: **The Other Side of Zero** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 19	Highest minus lowest.
2	**(d)** -15	Subtract the descent.
3	**(a)** -9 and 4	Check each pair.
4	**(a)** 7	Count the positive integers up to b-1.
5	**(c)** -3	Add step by step.
6	**(c)** 13	Add the opposites.
7	**(b)** 0	Opposites cancel.
8	**(b)** 26	Highest minus lowest.
9	**(d)** -150	Subtract the descent.
10	**(b)** 15 and -8	Check each pair.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$14-(-5)=19$ (1); $-5-14=-19$ (1).	1+1
12	Integers: -6, -5, -4, -3, -2 (1); Number $=5$ (1).	1+1
13	Add the first two: -20 (1); Then add 28: 8 (1).	1+1
14	$-12-(-7)=-5$ (1); $-7-(-12)=5$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Start at 0; after 4 down: -4; After 9 up: 5 (1); After 3 down: 2 (1); Total floors moved $=16$ (1).	1+1+1
16	After first: 150; After second: -150 (1); After deposit: 50 (1); Positive balance (1).	1+1+1
17	$10-(-6)=16$ (1); $16+12=28$ (1); $28-(-8)=36$ (1).	1+1+1
	OR: Start at 0; after 5 down: -5; After 1 up: -4 (1); After 6 down: -10 (1); Total floors moved $=12$ (1).	1+1+1
18	After first: -40; After second: -100 (1); After deposit: -10 (1); Negative means overdrawn (1).	1+1+1
	OR: $-5-(-9)=4$ (1); $4+4=8$ (1); $8-11=-3$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	After first jump: -1 (1); After second: -8 (1); After third: -11 (1); Distance from 0 $=11$ (2).	1+1+1+2
	OR: Coldest to hottest: -7, -3, -1, 5, 8 (1); Hottest 8, coldest -7 (1); Difference $=15$ (1); Sum $=2$ (2).	1+1+1+2
20	After first jump: 9 (1); After second: -3 (1); After third: -8 (1); Distance from 0 $=8$ (2).	1+1+1+2
	OR: Coldest to hottest: -15, -6, -4, 2, 9 (1); Hottest 9, coldest -15 (1); Difference $=24$ (1); Sum $=-14$ (2).	1+1+1+2

Mistakes to avoid:

- Saying $-7>-3$ because 7 is greater than 3; on the number line -7 lies to the left of -3.
- Writing $-4-(-6)=-10$; the correct value is $-4+6=2$.
- Dropping the sign of the bigger number when adding integers with different signs, e.g. $(-9)+5=4$ instead of -4.
- Taking the successor of a negative integer as one smaller (successor of -1 is 0, not -2).
- Finding a fall or rise as initial − final in the wrong order; use final − initial and read the sign.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 10: **The Other Side of Zero** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 10	Count the positive integers up to b-1.
2	**(b)** 2	Add step by step.
3	**(a)** -10	Add the opposites.
4	**(d)** 0	Opposites cancel.
5	**(b)** 12	Highest minus lowest.
6	**(d)** -70	Subtract the descent.
7	**(d)** -6 and -10	Check each pair.
8	**(a)** 5	Count the positive integers up to b-1.
9	**(c)** -5	Add step by step.
10	**(c)** -2	Add the opposites.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Integers: -2, -1, 0, 1, 2, 3 (1); Number $= 6$ (1).	1+1
12	Add the first two: -14 (1); Then add 7: -7 (1).	1+1
13	$8 - (-11) = 19$ (1); $-11 - 8 = -19$ (1).	1+1
14	Integers: -8, -7, -6, -5 (1); Number $= 4$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Start at 0; after 2 down: -2; After 8 up: 6 (1); After 5 down: 1 (1); Total floors moved $= 15$ (1).	1+1+1
16	After first: 110; After second: -50 (1); After deposit: -10 (1); Negative means overdrawn (1).	1+1+1
17	$12 - 15 = -3$ (1); $-3 + (-20) = -23$ (1); $-23 - (-6) = -17$ (1).	1+1+1
	OR: Start at 0; after 7 down: -7; After 3 up: -4 (1); After 9 down: -13 (1); Total floors moved $= 19$ (1).	1+1+1
18	After first: 125; After second: -175 (1); After deposit: -25 (1); Negative means overdrawn (1).	1+1+1
	OR: $-3 - 7 = -10$ (1); $-10 + (-11) = -21$ (1); $-21 - 4 = -25$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	After first jump: 7 (1); After second: 1 (1); After third: -13 (1); Distance from 0 $= 13$ (2).	1+1+1+2
	OR: Coldest to hottest: -10, -8, -1, 7, 13 (1); Hottest 13, coldest -10 (1); Difference $= 23$ (1); Sum $= 1$ (2).	1+1+1+2
20	After first jump: 7 (1); After second: -4 (1); After third: -12 (1); Distance from 0 $= 12$ (2).	1+1+1+2
	OR: Coldest to hottest: -11, -6, -2, 4, 10 (1); Hottest 10, coldest -11 (1); Difference $= 21$ (1); Sum $= -5$ (2).	1+1+1+2

Mistakes to avoid:

- Saying $-7 > -3$ because 7 is greater than 3; on the number line -7 lies to the left of -3.
- Writing $-4 - (-6) = -10$; the correct value is $-4 + 6 = 2$.
- Dropping the sign of the bigger number when adding integers with different signs, e.g. $(-9) + 5 = 4$ instead of -4.
- Taking the successor of a negative integer as one smaller (successor of -1 is 0, not -2).
- Finding a fall or rise as initial − final in the wrong order; use final − initial and read the sign.

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